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ABSTRACT

The water-dropping of firefighting aircraft is a complex gas–liquid two-phase process generating multi-scale droplets, and its accurate simulation is vital for assessing fire-extinguishing performance. Mesh-based methods (Volume of Fluid, VOF) capture interfaces accurately but are computationally expensive for large-scale problems, whereas meshless methods, such as the discrete phase model (DPM), efficiently simulate droplet motion but fail to describe breakup dynamics. This study proposes an enhanced DPM that incorporates a Weber number-based breakup criterion, allowing droplet splitting and a more realistic splashing evolution. The model is validated against water jet experiments, which show high consistency in liquid column formation and droplet breakup. A VOF-enhanced-DPM algorithm is further established to simulate the overall water-dropping morphology of firefighting aircraft, and the enhanced-DPM is independently applied to analyze water landing distribution under varying discharge volumes, heights and velocities, and to compare the effective fire-extinguishing range. Results demonstrate that the VOF-enhanced-DPM algorithm provides reliable precision simulations. In addition, the enhanced-DPM achieves a balance between computational efficiency and predictive accuracy in large-scale water-dropping applications.

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I. INTRODUCTION

Forest wildfires are among the most destructive natural disasters worldwide, posing severe threats to ecosystems, biodiversity, and human safety.^{1,2} In recent years, driven by climate change, the frequency and intensity of wildfires have significantly increased in many regions.³ Firefighting aircraft, with advantages of rapid response, wide operational range, and high safety, have become a crucial tool for extinguishing large-scale forest fires and are widely used in many countries.^{4–6} Despite their extensive application, the aerial water-dropping process involves complex physical phenomena—including droplet deformation, breakup, coalescence, and atomization, which makes accurate prediction of droplet behavior highly challenging.^{7–11} Accurate numerical simulation has therefore become a critical research focus.

With the continuous advancement of Computational Fluid Dynamics (CFD), an increasing number of studies have investigated

the dynamic characteristics of aerial water-dropping using numerical approaches.¹² In 2004, the Fire and Disaster Management Agency of Japan¹³ developed the Nagare Code, a finite difference solver of the Navier–Stokes (N–S) equations providing theoretical support for liquid spraying simulations. Satoh *et al.*¹⁴ validated the reliability of numerical simulation in predicting the water-spray patterns by comparing three-dimensional simulations with field tests. Ito *et al.*¹⁵ examined the global evolution of liquid after deployment through wind tunnel experiments and proposed using the volume of fluid (VOF) method to simulate water deformation, while applying the discrete phase model (DPM) to calculate droplet motion after breakup. However, this proposal was validated only for small-scale aircraft models. Borisov and Tsipenko¹⁶ used FlowVision software to simulate the water-dropping process of the Ka-32 helicopter, solving the compressible Navier–Stokes (N–S) equations with a turbulence model and incorporating vaporization effects. Zhao *et al.*¹⁷ applied the Eulerian

multiphase flow model in FLUENT to obtain the groundwater-landing density distribution, with $\sim 30\%$ error in the maximum density but reasonable agreement in overall distribution. Ren *et al.*¹⁸ used the VOF method for direct numerical simulation of water droplets in turbulent flow. Zhou *et al.*¹⁹ improved the simulation accuracy of the water-dropping process by employing the VOF method with mesh adaptation technology to analyze helicopter tank discharges. Rouaix *et al.*²⁰ numerically investigated the near-nozzle behavior of circular liquid jets through large-diameter nozzles using the VOF method. In the same year, Calbrix *et al.*²¹ applied the VOF method to study droplet breakup and dispersion from Canadair CL-415 and Dash-8 aircraft, gaining comprehensive insights into the liquid evolution during free fall. Guan *et al.*²² employed an improved Smoothed Particle Hydrodynamics (SPH) method, achieving good agreement in overall water-dropping morphology but with relatively large error in water-dropping density. Sun *et al.*²³ proposed a customized decoupled algorithm to calculate groundwater landing distribution, verified through DC-10 drop tests and low-velocity wind tunnel experiments, and further analyzed the effects of drop height, ground velocity, discharge mass flow rate, and crosswind on fire-extinguishing efficiency. Li *et al.*²⁴ conducted a numerical simulation of the aerial water-dropping process using the Discrete Element Method (DEM). The study revealed that the initial release velocity is the key factor determining the width of the water distribution, while both the initial velocity and the water volume jointly influence its longitudinal extent. Sun *et al.*²⁵ developed a mathematical model to calculate the trajectory and impact patterns of fire-extinguishing bags. The results indicated that the effectiveness of the bags is not significantly affected by the drop height, and acceptable drop patterns can still be achieved under high-altitude dropping conditions.

Beyond the specific domain of aerial firefighting, the broader computational fluid dynamics community has developed a rich literature on handling complex multiphase flows. To address the conflict between resolution and efficiency, adaptive particle methods have attracted increasing attention. For instance, Vacondio *et al.*²⁶ proposed a dynamic particle coalescing and splitting scheme guided by a background grid to optimize resolution, while Chen *et al.*²⁷ developed an adaptive multi-resolution smoothed particle hydrodynamics (SPH) method for three-dimensional free-surface flows to improve computational efficiency while maintaining accuracy in impact and free-surface regions. More recently, Yang *et al.*²⁸ integrated adaptive particle refinement with GPU (Graphics Processing Unit) acceleration technology to simulate complex multi-phase flows, significantly reducing computational costs. In addition, Koch *et al.*²⁹ applied SPH to simulate the primary atomization of atomizers, demonstrating the method's capability in capturing complex ligament breakup and droplet formation without mesh constraints. Similarly, coupled particle-mesh solvers have been investigated to combine the flexibility of Lagrangian particles with the efficiency of Eulerian grids.³⁰ Furthermore, in fields such as fuel atomization^{31,32} and agricultural aviation,^{33,34} hybrid strategies coupling VOF with DPM have proven to be powerful tools for capturing multiscale spray dynamics. These advanced methodologies provide a solid theoretical foundation for improving aerial water-dropping simulations.

Although these aforementioned approaches have made significant progress, applying them directly to the massive scale of aerial firefighting remains challenging, and existing methods still exhibit obvious limitations.³⁵ On the one hand, Eulerian approaches require

extremely fine grids to resolve the evolution of tiny droplets in large-scale or complex flow fields. While adaptive mesh refinement (AMR) techniques can partially alleviate this burden, the multi-scale nature of aerial water-dropping—spanning from a large flight trajectory (hundreds of meters) to millimeter-sized droplets—renders interface-resolving simulations prohibitively expensive. This leads to high computational cost, low efficiency, and considerable challenges in statistically analyzing droplet distributions or calculating ground coverage density. On the other hand, Lagrangian or hybrid methods also face constraints. Although adaptive particle methods and coupled particle-mesh solvers have been proposed to improve efficiency, they often introduce numerical noise during particle splitting or rely on complex interface coupling algorithms, which are prone to interpolation errors and stability issues in violent breakup scenarios. Furthermore, because the initial particle size remains fixed in traditional discrete phase calculations, these methods cannot dynamically split droplets according to physical breakup processes, often introducing substantial errors, particularly in predicting droplet breakup and spatial distribution.

Given these limitations, this study proposes an improved discrete phase model (enhanced-DPM). This model introduces dynamic breakup and splitting of droplets based on the Weber number criterion, thereby effectively overcoming the accuracy limitations of fixed particle size in conventional discrete phase methods. To balance computational accuracy and efficiency between the continuous and discrete phases, the enhanced-DPM is coupled with the VOF method. The VOF method is employed to capture the initial morphology of the liquid column and the evolution of the gas-liquid interface. At the same time, the enhanced-DPM efficiently simulates droplet breakup, dispersion, and deposition. Finally, the accuracy of the proposed model in predicting far-field droplet behavior is subsequently validated.

This paper is organized as follows: Section II presents the theoretical foundations and numerical methods, including the VOF model, the proposed enhanced-DPM, and its coupling strategy; Sec. III introduces the experimental validation of the enhanced-DPM and demonstrates the effectiveness of the coupling strategy using a small-scale experimental case; Sec. IV employs the enhanced-DPM to analyze the effects of discharge volumes, drop heights, and flight velocity on water landing distribution, and further evaluates the fire-extinguishing performance under different conditions; Sec. V summarizes the results and outlines potential directions for future research.

II. NUMERICAL SIMULATION METHOD

Numerical simulation of the water-dropping process by firefighting aircraft requires accurate characterization of the evolution of the liquid column in high-velocity airflow after ejection from the nozzle. Furthermore, the multiscale dynamics of column breakup, droplet fragmentation, and gas-liquid interaction must be considered. This process involves the transformation of a continuous liquid column into dispersed droplets, manifested through severe free-surface deformation, breakup of large droplets, and atomization and settling of smaller droplets. A coupled VOF-enhanced-DPM algorithm is developed to capture both interface evolution and droplet motion.

A. Volume of fluid (VOF) method

The VOF method³⁶ is a typical Euler-Euler description method that introduces a volume fraction function to characterize the proportion of gas and liquid phases in the same control volume

$$\alpha = \frac{V_l}{V}, \quad (1)$$

where V_l is the liquid volume in the cell, V is the total cell volume. For any computational cell, if $\alpha = 1$, it means the cell is filled with liquid, $\alpha = 0$ means it is filled with air, and $0 < \alpha < 1$ indicates the cell is at the gas–liquid interface.

The transport equation for volume fraction can be expressed as

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \mathbf{u}) = 0, \quad (2)$$

where α is the volume fraction, and $\mathbf{u}$ is the velocity vector. The VOF method captures the gas–liquid interface by updating the volume fraction field.

In terms of momentum conservation, the VOF method solves the incompressible Navier–Stokes equations

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \nabla \cdot (\mu \nabla \mathbf{u}) + \rho \mathbf{g} + \mathbf{F}_\sigma, \quad (3)$$

where ρ and μ are the density and viscosity of the air–water mixture, respectively, and $\mathbf{F}_\sigma$ represents the surface tension force. The surface tension is calculated using the continuum surface force (CSF) model proposed by Brackbill *et al.*³⁷ which is calculated as

$$\mathbf{F}_\sigma = \sigma \kappa \nabla F, \quad (4)$$

where σ is the surface tension coefficient and κ is the curvature. The mixture properties are weighted by volume fraction

$$\rho = \alpha \rho_l + (1 - \alpha) \rho_g, \quad \mu = \alpha \mu_l + (1 - \alpha) \mu_g. \quad (5)$$

The gas–liquid interface is reconstructed using a geometric reconstruction piecewise-linear interface calculation (PLIC) scheme,³⁸ which ensures a sharp representation of the interface and improves numerical accuracy compared with algebraic VOF schemes.

The turbulence effects are modeled using the Shear Stress Transport (SST) k – ω model,³⁹ which solves transport equations for the turbulent kinetic energy k and the specific dissipation rate ω

$$\frac{\partial(\rho k)}{\partial t} + \nabla \cdot (\rho \mathbf{u} k) = P_k - \beta^* \rho k \omega + \nabla \cdot [(\mu + \mu_t / \sigma_k) \nabla k], \quad (6)$$

$$\frac{\partial(\rho \omega)}{\partial t} + \nabla \cdot (\rho \mathbf{u} \omega) = \alpha \frac{\omega}{k} P_k - \beta \rho \omega^2 + \nabla \cdot [(\mu + \mu_t / \sigma_\omega) \nabla \omega], \quad (7)$$

where P_k is the production term of turbulent kinetic energy, $\mathbf{u}$ is velocity vector, α , β , and β^* are model constants, σ_k and σ_ω are turbulent Prandtl numbers.

B. Enhanced-discrete phase model (enhanced-DPM)

After the liquid column evolution is resolved using the VOF method, the detached droplets are converted into Lagrangian particles and subsequently modeled using the Discrete Phase Model (DPM).⁴⁰ The DPM is a hybrid Eulerian–Lagrangian method in which the gas phase is solved as a continuous medium using the Navier–Stokes equations, and droplets are tracked as Lagrangian particles. The governing equation of motion for a single droplet is

$$m_p \frac{d\mathbf{u}_p}{dt} = \mathbf{F}_D + m_p \mathbf{g} + \mathbf{F}_{add}, \quad (8)$$

where m_p is the droplet mass, $\mathbf{u}_p$ is the droplet velocity, $\mathbf{F}_D$ is the aerodynamic drag, $m_p \mathbf{g}$ is the gravitational force, and $\mathbf{F}_{add}$ represents other additional forces. The drag force is commonly expressed as

$$\mathbf{F}_D = \frac{1}{2} C_D \rho_g A_p |\mathbf{u} - \mathbf{u}_p| (\mathbf{u} - \mathbf{u}_p), \quad (9)$$

where C_D is the drag coefficient, ρ_g is the air density, and A_p is the droplet frontal area.

The limitation of traditional DPM is that the droplet size remains fixed throughout the calculation. Once the initial particle size is set, no dynamic splitting occurs, failing to reproduce the physical breakup process. To address this shortcoming, the enhanced-DPM has been developed, incorporating the following improvements:

1. Weber number breakup criterion

The Weber number (We) is an important dimensionless parameter for determining droplet breakup, given by

$$We = \frac{\rho_g U_r^2 d_p}{\sigma}, \quad (10)$$

where ρ_g is the air density, U_r is the relative velocity between the droplet and airflow, d_p is the droplet diameter, and σ is the surface tension. As shown in Fig. 1, droplets breakup when We exceeds the critical value of 12.

2. Dynamic splitting algorithm

Droplets that satisfy the breakup criterion are designated as parent particles, which split into multiple child particles (Fig. 2). The diameters of child droplets are constrained by mass conservation

$$\sum_i d_{c,i}^3 = d_p^3, \quad (11)$$

where d_p is the diameter of the parent droplet, and d_c is the diameter of the child droplet. The velocity of each child droplet is determined by the parent velocity and a normal perturbation velocity

$$\mathbf{u}_{c,i} = \mathbf{u}_p + \mathbf{u}_n, \quad (12)$$

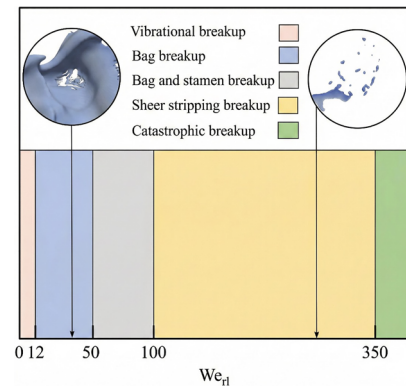


FIG. 1. Schematic of droplet breakup dependence on Weber number.⁴¹ Reproduced with permission from Sun *et al.*, *Int. J. Multiphase Flow* **170**, 104655 (2024), Copyright 2024 Elsevier.

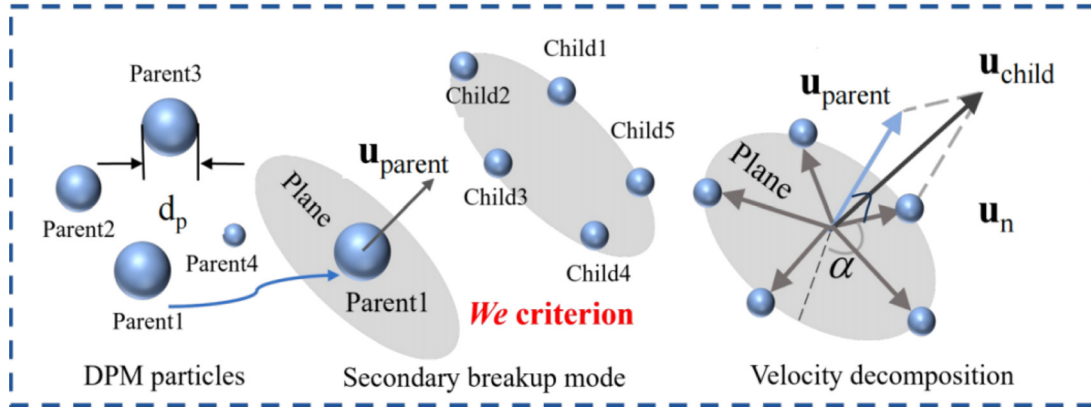


FIG. 2. Droplet breakup model for the enhanced-DPM.

where u_p is the parent droplet velocity, and u_n is the perturbation velocity perpendicular to the parent droplet velocity direction. The direction α of u_n is randomly selected within $[0, 2\pi]$, and its magnitude is given by

$$u_n = n \frac{d_p}{t_b - t_{def}}, \quad (13)$$

where t_b is the droplet deformation period and t_{def} is the droplet breakup characteristic time. The calculation formula is as follows:

$$t^* = \frac{d_{c,i}}{U_r} \sqrt{\frac{\rho_l}{\rho_g}}, \quad (14)$$

where U_r is the relative velocity between droplet and gas. The calculation expression of t_b is as follows:⁴²

$$\frac{t_b}{t^*} = \begin{cases} 6(We - 12)^{-0.25} & 12 < We \leq 18, \\ 2.45(We - 12)^{0.25} & 18 < We \leq 45, \\ 14.1(We - 12)^{-0.25} & 45 < We \leq 351, \\ 0.776(We - 12)^{0.25} & 351 < We \leq 2670, \\ 5.5 & 2670 < We. \end{cases} \quad (15)$$

The pseudo-code for the dynamic droplet breakup based on the Weber number is given in Algorithm 1.

C. VOF-enhanced-DPM coupling strategy

The water-dropping process involves both the continuous evolution of the liquid column and the dispersed motion of droplets. To balance accuracy and efficiency, the VOF method is employed in the initial phase to resolve large-scale deformation and breakup of the liquid column. The separated droplets are then converted into DPM particles for tracking. The conversion is governed by droplet equivalent diameter and asphericity³¹

$$d_{eq} \leq d_{thr}, \quad \xi \leq \xi_c, \quad (16)$$

where d_{eq} is the equivalent diameter, d_{thr} is the threshold diameter, ξ is the asphericity, and ξ_c is the critical value.

During conversion, the liquid volume is removed from the VOF solver and mapped into the Lagrangian framework of the enhanced-DPM, ensuring conservation of mass, momentum, and position.

D. Fire-extinguishing evaluation criteria

1. Empirical criteria

Fire-extinguishing performance is often evaluated using the water landing density (the total volume of water per unit area) as a key indicator. According to Yamashita *et al.*,⁴³ a density of $\sim 2 \text{ l/m}^2$ is generally required for effective fire extinguishing. In contrast, water drop tests

ALGORITHM 1: Dynamic droplet breakup based on Weber number.

Input:

1. d_p, u_p, x_p : Diameter, velocity, and position of parent droplet.
2. ρ_g, σ, U_r : Gas density, surface tension, and relative velocity.
3. $We_{critical}, \Delta t$: Critical Weber number for breakup and time step.

Output: d_c, u_c, x_c : Diameter, velocity, and position for child droplets.

Initialize the candidate pool from VOF data

while P is not empty **do**

Pop a droplet from P

Calculate Weber number: $We = \rho_g U_r^2 d_p / \sigma$

if $We > 12$ **then**

Calculate t_{def} and t_b according to Eqs. (14) and (15)

Calculate $d_c = d_p / N^{1/3}$ Mass conservation

Calculate perturbation magnitude: $u_n = nd_p / (t_b - t_{def})$

for each child droplet $j = 1$ to N **do**

Sample random direction $\alpha \in [0, 2\pi]$, where $u_n \perp u_p$

Update child velocity: $u_{c,i} = u_p + u_n$ back into P for evaluation

Update child position: $x_{c,i} = x_p + u_{c,i} \Delta t$

end

end

End

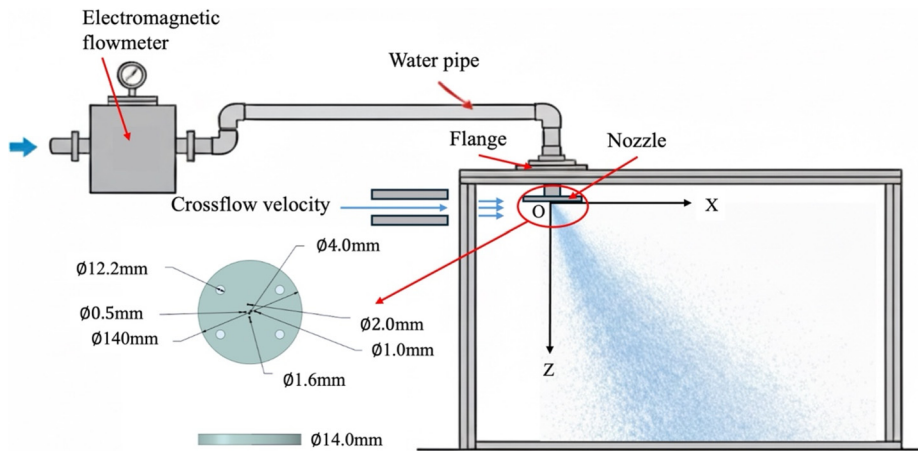


FIG. 3. Schematic diagram of the transverse jet test platform.

conducted in Europe and the United States commonly adopt 0.2 l/m^2 as the threshold to determine the effective coverage area of falling water.¹⁰

2. Heat-exchange model

Traditional fire-extinguishing assessments mainly rely on empirical thresholds, which provide only limited evaluation bases. To achieve a more rigorous and flexible evaluation, this study develops a heat-exchange model that considers both direct heat absorption by the extinguishing agent and heat removal through evaporation.

The total absorbed heat includes sensible and latent heat as shown in the following equation:

$$Q_i = m \cdot C_p \cdot \Delta T + m \cdot L_v, \quad (17)$$

where Q_i is the heat absorbed by water, m is water mass, C_p is the specific heat capacity of water, typically $4.18 \text{ J/g} \cdot ^\circ\text{C}$, ΔT is the temperature change, and L_v is the latent heat of vaporization ($2.26 \times 10^6 \text{ J/g}$). The heat of a fire can be calculated using the following formula:

$$Q_h = H_{\text{release}} \cdot t, \quad (18)$$

where H_{release} is the heat release rate of the fire source, and t is the burning duration. The heat release rate depends on the combustion characteristics of the fuel and environmental conditions, and can be estimated as

$$H_{\text{release}} = \varepsilon \cdot \Delta H_{\text{combustion}}, \quad (19)$$

where ε is the burning rate, and $\Delta H_{\text{combustion}}$ is the heat of combustion.

In the case of wood fires, extinguishing effectiveness is strongly linked to temperature reduction. Studies have shown that when the flame temperature (typically $600\text{--}1000^\circ\text{C}$) decreases, and the wood surface temperature drops below 100°C , combustion is no longer sustained, and the fire can be considered extinguished.^{44–46} Thus, the critical temperature drop serves as a key criterion for evaluating fire-extinguishing effectiveness. It is important to note that while Eq. (17) accounts for the latent heat of vaporization to evaluate the total cooling capacity, the mass reduction of droplets due to evaporation during their flight through the high-temperature zone is not explicitly

modeled in the current DPM simulations. This simplification focuses on the aerodynamic transport of the extinguishing agent, providing a conservative estimate of the water flux reaching the fire source by isolating the trajectory analysis from the complex and highly uncertain evaporation rates during the flight phase.

III. MODEL VALIDATION

Traditional DPM often assumes fixed droplet sizes and neglects breakup under strong transverse airflow, which causes significant deviations from experiments. To overcome this limitation, this study developed an enhanced-DPM. Its prediction accuracy is validated against transverse jet experiments.

A. Enhanced-DPM experimental comparison and verification

The experimental setup (Fig. 3) consists of a $4.0 \pm 0.02 \text{ mm}$ diameter nozzle ejecting water into a transverse airflow generated by a blower. The jet behavior was captured with a high-speed camera (Phantom V711) operating at 5000 Hz with an exposure time of $2 \mu\text{s}$.

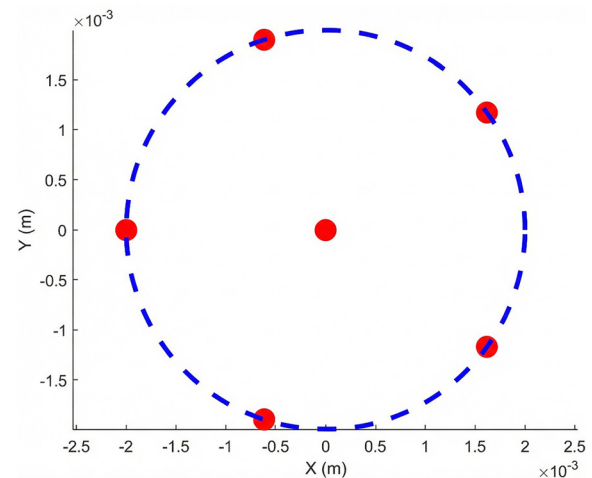


FIG. 4. Numerical model of particle nozzle.

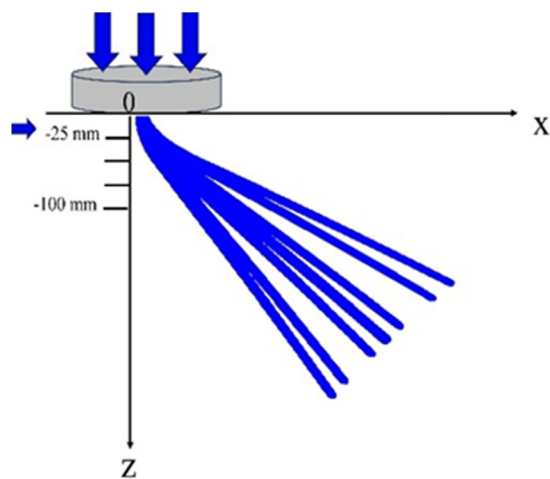


FIG. 5. Wind velocity range setting.

In the numerical model, the jet was aligned with the z axis, and the airflow with the x axis. The initial droplet equivalent diameter was 2 mm matched the nozzle flow rate. Six representative particles were introduced at the nozzle: five evenly distributed along the nozzle circumference, and one was at the center, ensuring uniformity. The nozzle particle model and airflow loading method are shown in Figs. 4 and 5.

The front angle of the liquid column was used for model verification, as shown in Figs. 6 and 7. At a wind velocity of 26 m/s, the traditional DPM without breakup overestimates the front angle. In contrast, the enhanced-DPM, incorporating the breakup criterion, closely reproduces liquid column deflection and spread.

B. VOF-enhanced-DPM numerical example verification

To verify the reliability of the VOF-enhanced-DPM coupling framework, a benchmark transverse-jet case from Li and Soteriou⁴⁷ was reproduced. Such experiments are widely used for droplet breakup validation due to their repeatability and comprehensive data.

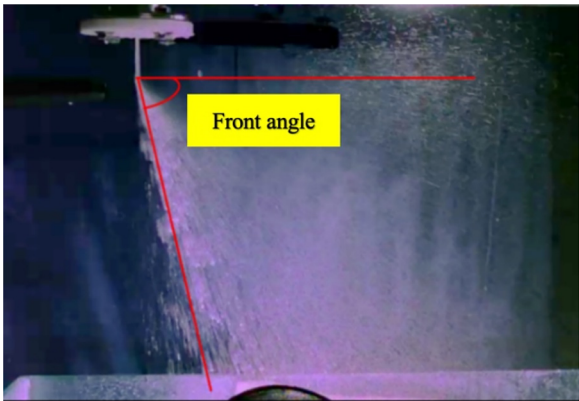


FIG. 6. Definition of front angle in transverse.

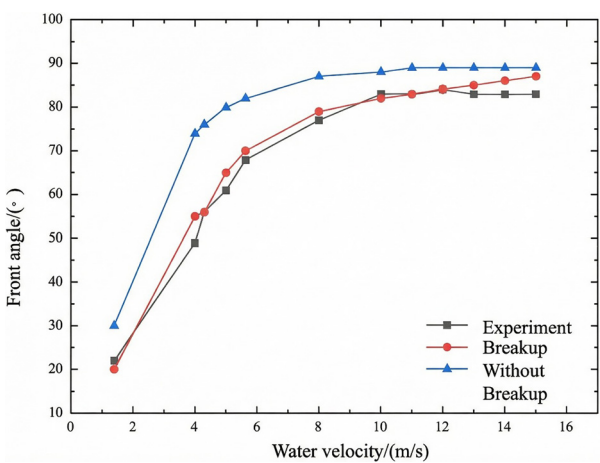


FIG. 7. Comparison of front angle at wind velocity of 26 m/s.

The computational domain is defined as $30 \times 20 \times 50 \text{ mm}^3$ (length, width, and height). The inlet coordinates are (2, 10, and 10), and the inlet diameter is 0.8 mm, as illustrated in Fig. 8. The $x = 0$ plane is a velocity inlet, the $y = 50$ plane is a no-slip wall, and all remaining boundaries are free outflow. Test conditions include a water velocity of 17.7 m/s, an air velocity of 54.8 m/s, a fixed momentum ratio of 88.2, and a Weber number $We = 40$. Material parameters are shown in Table I. A mesh independence study was performed to assess the sensitivity of the numerical results to mesh resolution. Three levels of adaptive mesh refinement

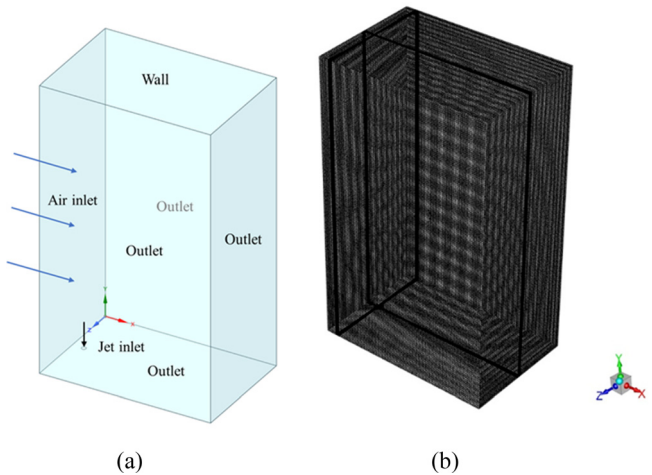


FIG. 8. Computational domain for transverse jet simulation: (a) geometric model and (b) meshing model.

TABLE I. Material properties.

Properties	$P \text{ (kg/m}^3\text{)}$	$M \text{ (Pa s)}$	$\sigma \text{ (N/s)}$
Air	1.18	1.86×10^{-6}	...
Water	997	8.94×10^{-4}	0.0708

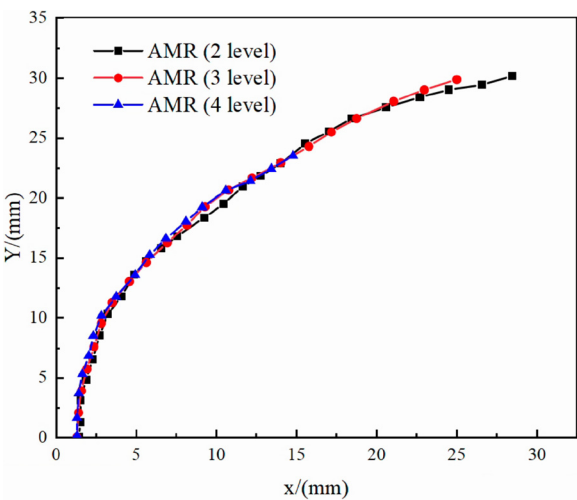


FIG. 9. Jet boundary for different AMR levels.

(AMR) were considered, corresponding to two, three, and four refinement levels (Fig. 9), respectively. It can be seen that the boundary changes are stable. With the three-level adaptive mesh, the minimum cell size reaches $38\text{ }\mu\text{m}$, which is comparable to and slightly finer than the $39\text{ }\mu\text{m}$ resolution reported in the literature. Based on this analysis, the mesh with three refinement levels was adopted for all subsequent simulations as a compromise between computational cost and accuracy. The simulated

TABLE II. Solver setting parameters.

Time	Transient
Multi-phase model	VOF-to-DPM
VOF model	Explicit; geo-reconstruct
Momentum discretization method	Second-order upwind
Turbulence	SST- $k-\omega$
Shear condition	No slip
Surface tension	Continuum surface force
Time step/s	1×10^{-7}

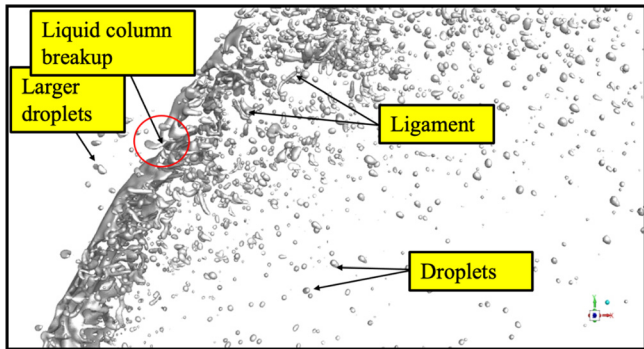


FIG. 10. VOF simulation results of transverse jet.

full penetration time is $\sim 2\text{ ms}$. The time step ranges from 10^{-8} to 10^{-5} s until residuals fall below 10^{-3} . The simulations were executed on an Intel(R) Xeon(R) Platinum 8352Y Central Processing Unit (CPU) server with 12 cores and 24 threads, with a frequency of 2.8 GHz and 64 GB of RAM. The use of a workstation for this class of problems is associated with speeding up the computing time. The physical simulation time was 0.1 s, corresponding to a computational time of approximately two weeks. Detailed solver settings are summarized in Table II.

1. Equivalent diameter and asphericity analysis

The conversion parameters, equivalent diameter, and asphericity were tested for sensitivity. As shown in Figs. 10 and 11, this parameter

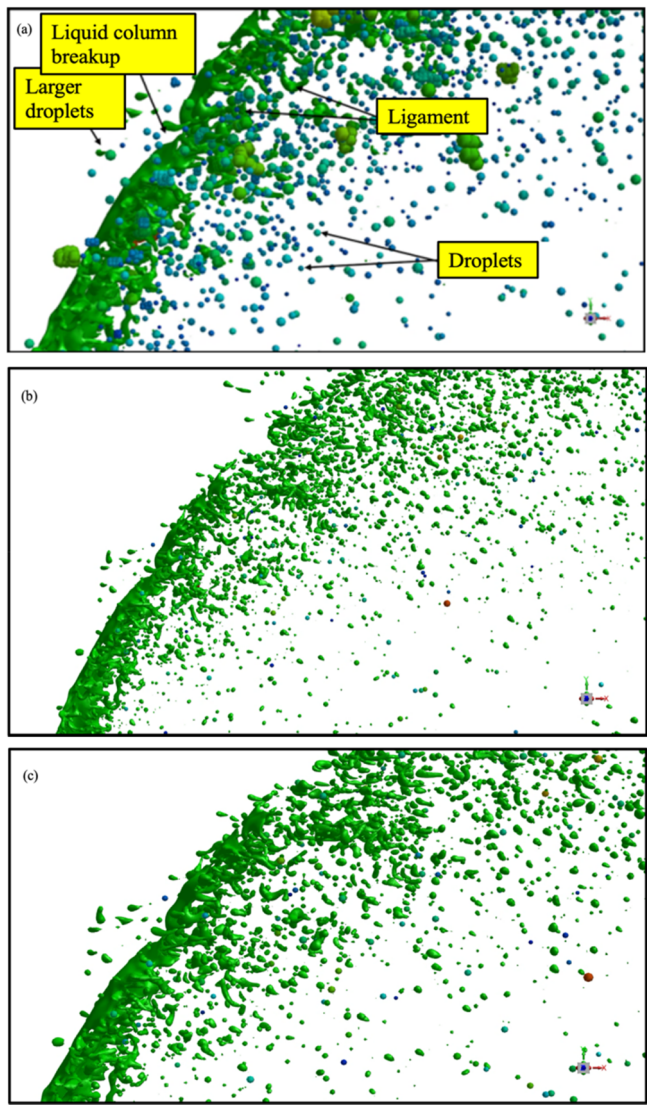


FIG. 11. Comparison of different conversion standards for transverse jet: (a) equivalent diameter is 0.4 mm, asphericity is 0.5; (b) equivalent diameter is 0.8 mm, asphericity is 0.5; and (c) equivalent diameter is 0.4 mm, asphericity is 0.1.

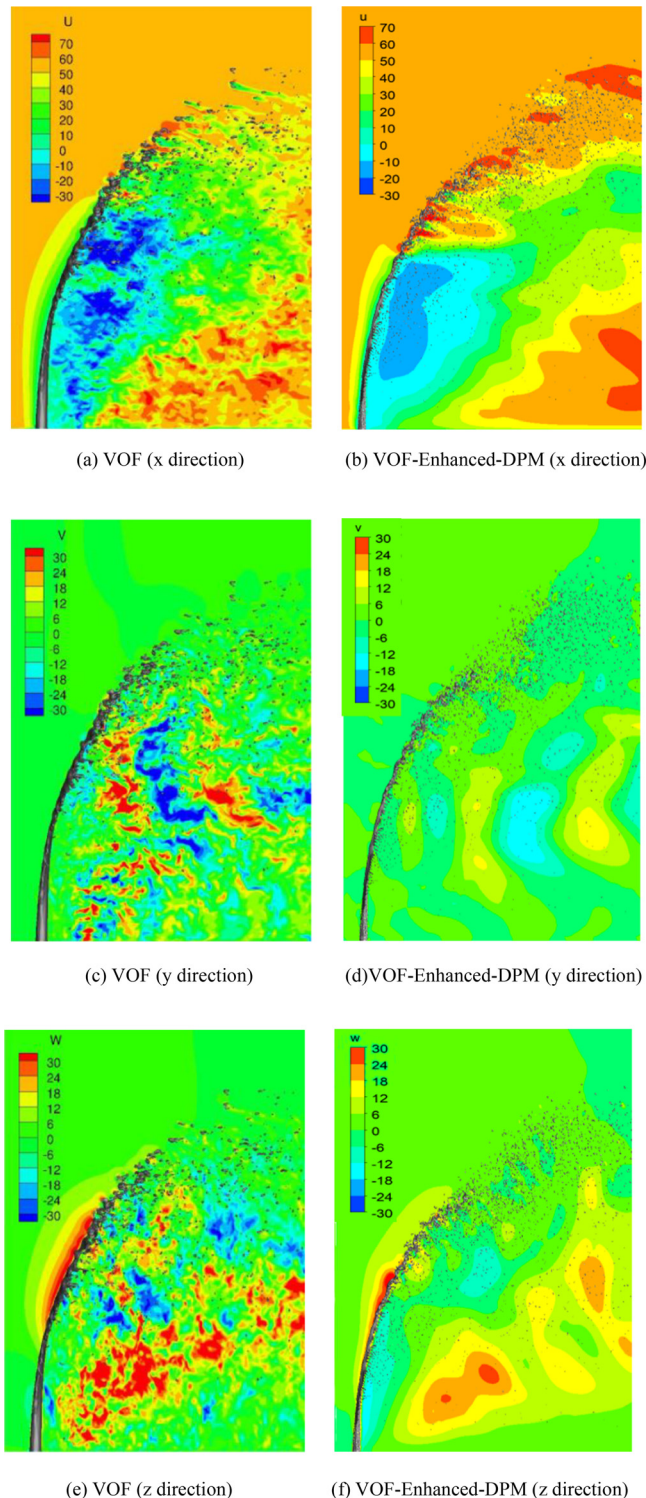


FIG. 12. Comparison of velocity cloud maps in the midplane ($y = 10$ mm).⁴⁷ Reproduced with permission from Phys. Fluids 28(8), 082101 (2016). Copyright 2016 AIP Publishing LLC.

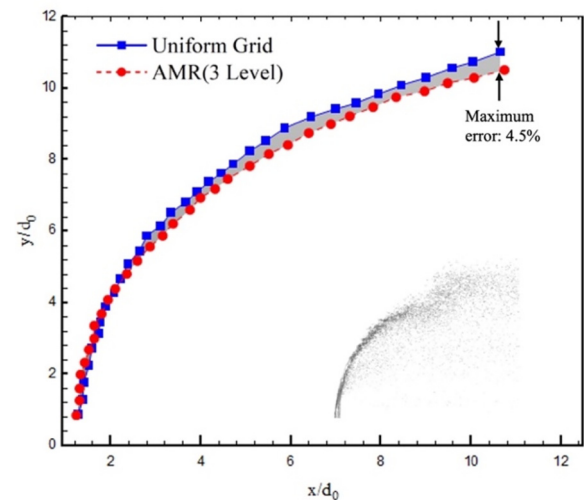


FIG. 13. Comparison of jet boundaries.

set best captured the ligament structures and ensured consistent phase conversion, while larger d_{eq} or smaller ζ values failed to reproduce the discrete droplets.

2. Velocity cloud maps analysis

To further assess the accuracy of the coupled algorithm, velocity contour maps from the present simulations were compared with the results of Li and Soteriou⁴⁷ in Fig. 12. The VOF-enhanced-DPM framework successfully reproduces the experimental phenomena, with predicted velocity distributions and jet expansion closely matching the reference data. These results demonstrate that the proposed coupling framework is reliable for simulating droplet breakup processes.

By extracting the left boundary of the jet from the literature and comparing it with the numerical simulation results of the present study, as shown in Fig. 13, the variation trend of the jet boundary can be clearly observed. The maximum error occurs at the highest point of the jet, with the maximum fitting error of the transverse jet boundary is 4.5%. This result demonstrates the effectiveness of the adaptive mesh technique in improving the accuracy of the numerical simulations.

3. Comparative analysis of computational efficiency and geometric fidelity

To quantitatively evaluate the applicability of the proposed Hybrid method, a benchmark study was conducted comparing the computational cost and prediction fidelity of VOF, Standard DPM, enhanced-DPM, and the Hybrid scheme. Table III summarizes the wall-clock time required for each method to simulate the same physical duration ($t = 0.1$ s). The full VOF simulation, serving as the high-fidelity baseline, employed a 3-level adaptive mesh refinement (AMR) to resolve the intricate interface structures. Consequently, it was computationally prohibitive, requiring ~ 500 h (about 3 weeks). In contrast, the Lagrangian-based methods (DPM and enhanced-DPM) demonstrated extremely high efficiency, completing the simulation in

TABLE III. Performance benchmark: computational cost vs accuracy.

Method	Physics resolution	Jet boundary deviation	Wall-clock time	Normalized	Applicability
VOF (adaptive)	Direct interface resolving	Baseline	~500 h (3 weeks)	100%	Validation/fundamental study
Enhanced-DPM	DPM + breakup model	Medium	~0.10 h	<0.05%	Quick estimation
Hybrid (VOF-enhanced-DPM)	VOF-enhanced-DPM)	Low (<15% deviation)	Depends on transition	~20%	Detailed engineering design
Standard DPM	Lagrangian point mass	High (>40% deviation)	~0.05 h	<0.01%	Far-field only

minutes. However, this speed comes at the cost of simplified physics, particularly in the near-nozzle region.

While computational speed is critical, it must not compromise the capture of essential flow features. Figure 14 compares the liquid jet boundaries predicted by the four methods. It is observed that the DPM fails to capture the continuous liquid core, resulting in a significant deviation in the predicted spray boundaries compared to the VOF baseline. Quantitative measurements of the spray boundaries indicate a deviation of over 50% for the pure Lagrangian methods. The proposed enhanced DPM and hybrid method reproduces the topology of the liquid column. By resolving the interface only in the primary breakup region and transitioning to DPM downstream, the Hybrid

method maintains a boundary prediction error of less than 15% relative to the VOF baseline.

C. VOF-enhanced-DPM engineering case verification

After validating the numerical method, the VOF-enhanced-DPM coupled framework was further applied to an aircraft water-dropping engineering case. Following the wind-tunnel test by Ito *et al.*¹⁵, a simplified fuselage-tank geometry (Figs. 15 and 16) was modeled to study water-dropping distribution, neglecting fuselage shape effects. The main parameters of the prototype and scaled model are summarized in Table IV.

Based on these working conditions, the simulation results of the VOF-enhanced-DPM coupling algorithm are shown in Fig. 17. Near the water inlet, the water flow is ejected as a continuum, and it gradually flattens under the shear of high-velocity airflow. In its downstream, the liquid column breaks up and forms ligaments and larger droplets. With the continued action of aerodynamic forces, the large droplets undergo atomization and break into smaller droplets. The VOF method captures the initial interface evolution and ligament formation, while the enhanced-DPM tracks subsequent droplet breakup and evolution.

Figure 18 shows the lateral boundaries of the experimental and simulated transverse jets. The shaded region indicates the approximate deviation between them. The corresponding fitting errors are also presented, with a maximum error of 20% under a velocity of 13.3 m/s and 10.6% under a velocity of 22.5 m/s. Overall, the simulation captures

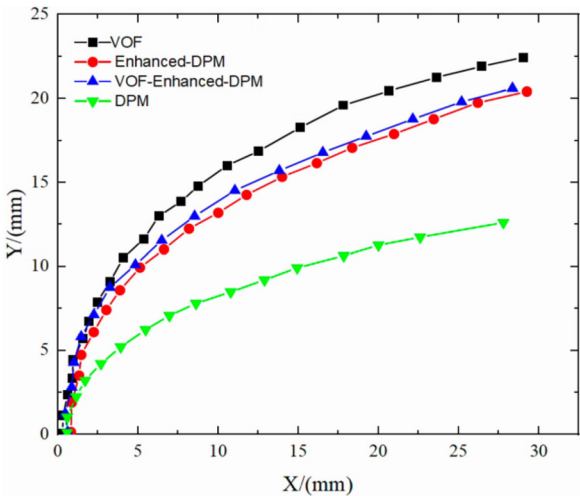


FIG. 14. Comparison of jet boundaries by different methods.

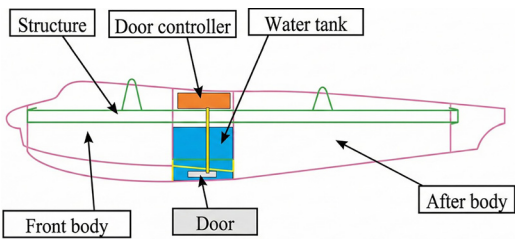


FIG. 15. Schematic diagram of the fuselage and water tank structure.

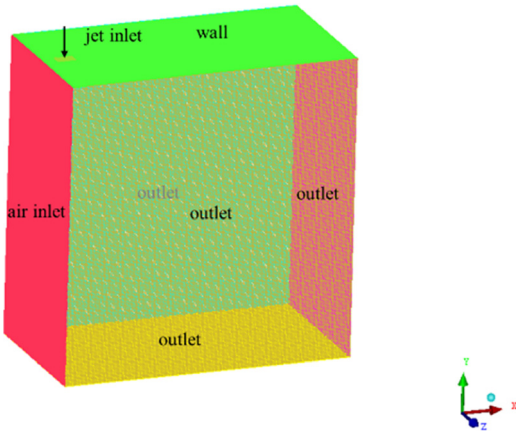


FIG. 16. Simplified geometric model for water-dropping simulation.

TABLE IV. Dimensions and working conditions of the prototype and scaled model.

Scale	Water volume (l)	Inlet size (mm)	Flight velocity (m/s)	Dropping height (m)
1	15	1920 × 640	45.2	30
1	15	1920 × 640	90	30
1/16	4.2	120 × 40	11.3	1.88
1/16	4.2	120 × 40	22.5	1.88

the jet spreading well, with minor deviations that are acceptable for the present analysis.

Although the VOF method is advantageous for capturing interface tracking, it is computationally expensive for large-scale problems with numerous droplets and wide size distributions. At the ground level, the liquid no longer exists as a continuum, and only the enhanced-DPM is applied to calculate the water-dropping distribution.

The model's engineering applicability was verified using scaled wind tunnel data from Ito *et al.*¹⁵ Figure 19 compares the measured and simulated water-dropping range. The enhanced-DPM predicted a water-dropping range of 4.5–9.0 m, with a maximum spread of ~0.5 m, and the overall distribution remained concentrated. A slight overestimation occurred in the region with water-film thickness above 3 mm. This discrepancy is primarily due to induced airflow, which reduces the relative velocity in the drag force calculation, and due to droplet deformation, which alters aerodynamic characteristics compared with the spherical assumption.

IV. RESULTS FOR THE AG600 FIREFIGHTING AIRCRAFT

A. Distribution characteristics of water-dropping parameters for the AG600 firefighting aircraft

After validation on small-scale cases, the enhanced-DPM method was applied to the AG600 firefighting aircraft as a representative large-scale example. The outlet dimensions are shown in Fig. 20, and the corresponding operating parameters are listed in Table V. The simulations analyzed the influence of water volume, drop height, and flight velocity on water distribution.

1. Influence of water volume

The distributions of water-dropping were calculated for water volumes of 3300, 6030, and 12 000 l, at a constant flight velocity of 230 km/h and a drop height of 50 m. The results are presented in Fig. 21. It was observed that with increasing water volume, the droplet density increased, and the falling water area became more concentrated.

2. Influence of drop height

Keeping the flight velocity constant at 230 km/h and the water volume at 12 000 l, simulations were performed for drop heights of 50, 75, 100, and 125 m. The results (Fig. 22) showed that the drop height expanded the coverage area but reduced droplet density and water-film thickness, including weaker local fire-extinguishing performance.

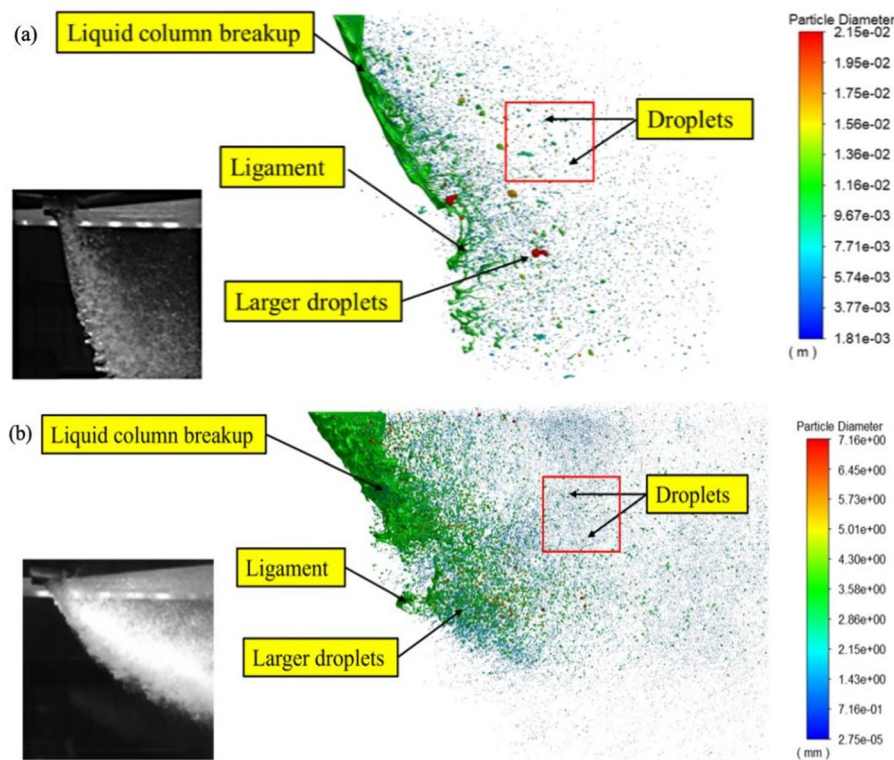
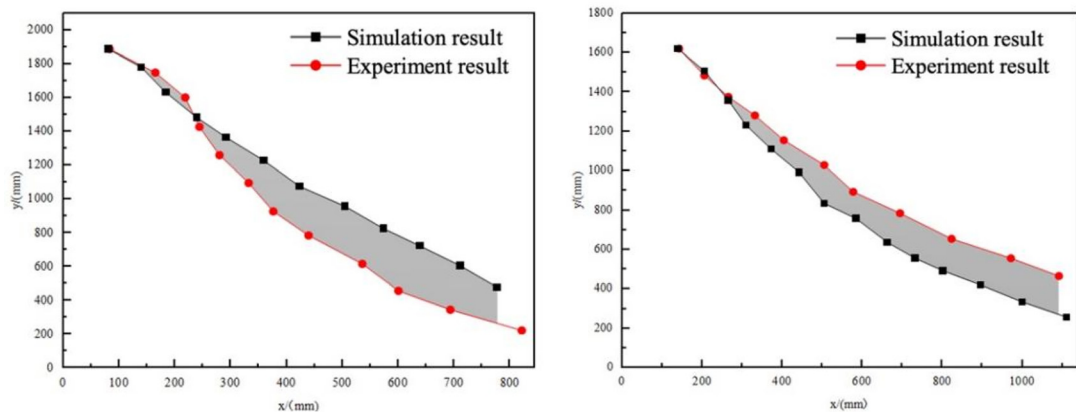


FIG. 17. Comparison of water-dropping results between experimental tunnel test and VOF-enhanced-DPM simulation: (a) wind velocity of 11.3 m/s and (b) wind velocity of 22.5 m/s.



(a) Jet boundary wind velocity of 11.3 m/s (b) Jet boundary wind velocity of 22.5 m/s

FIG. 18. Comparison of water-dropping boundary under different wind velocities.

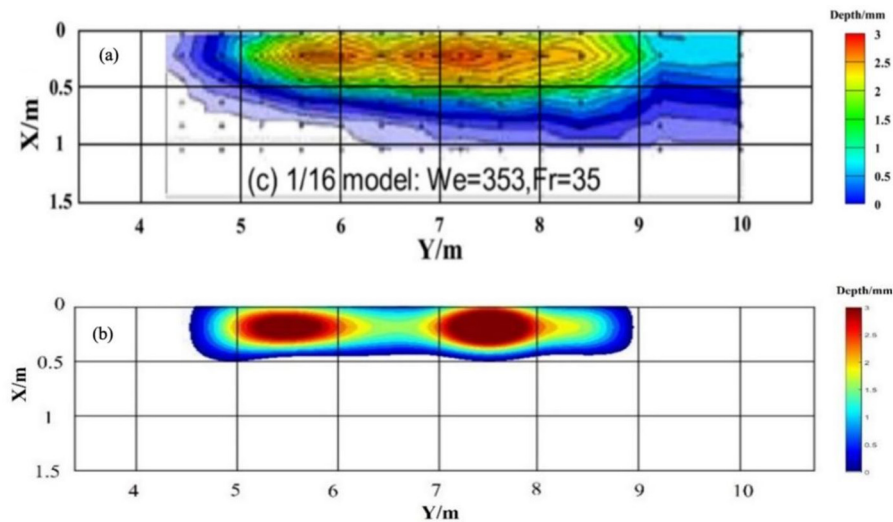


FIG. 19. Water-dropping range and density results (wind velocity 22.5 m/s, dropping height 1.88 m) (a) Ito *et al.* and (b) enhanced-DPM simulation.

3. Influence of flight velocity

For a water volume of 12 000 l and a drop height of 50 m, cases with flight velocities of 150, 200, 230, and 250 km/h were analyzed. The results (Fig. 23) demonstrated that with increasing flight velocity,

the falling water distribution shifted downstream, and the density distribution became less uniform. At lower velocity (150 km/h), the water was mainly concentrated beneath the aircraft, whereas at higher velocity (250 km/h), the distribution area broadened significantly.

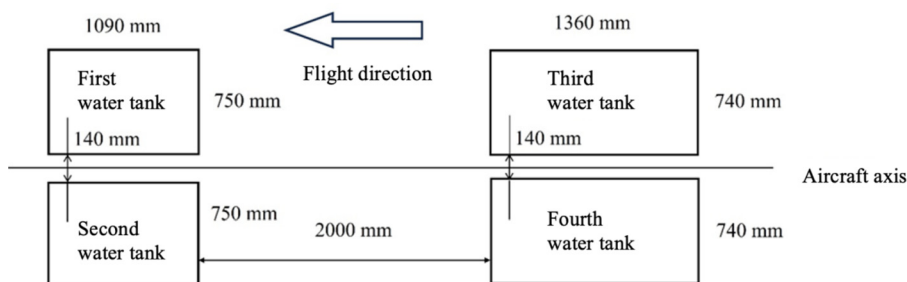


FIG. 20. Simplified position and size of water tank.

TABLE V. Operating parameters.

Case	Flight velocity (km/h)	Drop height (m)	Volume (l)	Time (s)	Outlet area (m ²)
1	230	50	3 330	1	1.01
2	230	50	6 030	1	1.82
3	230	50	12 000	1	3.65
4	230	75	12 000	1	3.65
5	230	100	12 000	1	3.65
6	230	125	12 000	1	3.65
7	150	50	12 000	1	3.65
8	200	50	12 000	1	3.65
9	250	50	12 000	1	3.65

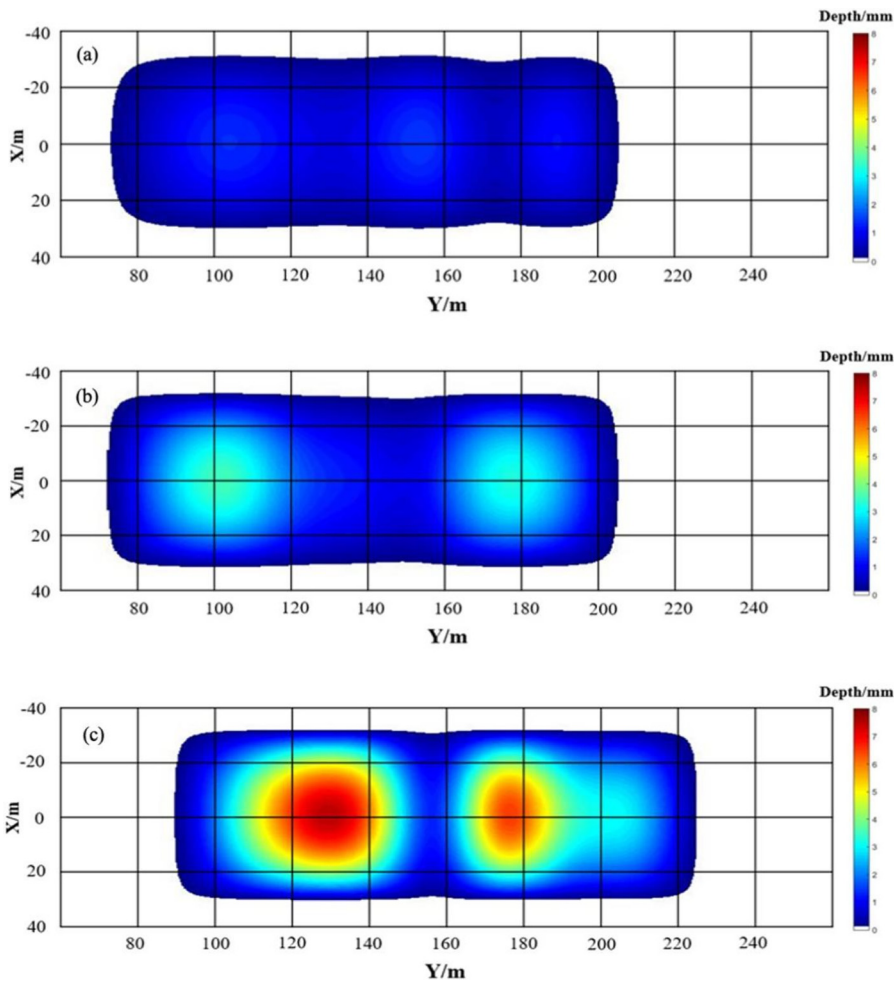


FIG. 21. Water-dropping range and density result: (a) water volume 3300 l; (b) water volume 6030 l; and (c) water volume 12 000 l.

B. Comprehensive evaluation of AG600 fire-extinguishing performance

Based on the heat-exchange model described in Sec. IID, the extinguishing performance of the AG600 aircraft was further evaluated. In this paper, wood was selected as the representative

combustible material, the initial fire source temperature was set to 600 °C, and the fire duration was fixed at 1 s. The fire is assumed to be extinguished once the surface temperature of the wood decreases to 100 °C. The average combustion calorific value of wood was taken as 20 kJ/g.

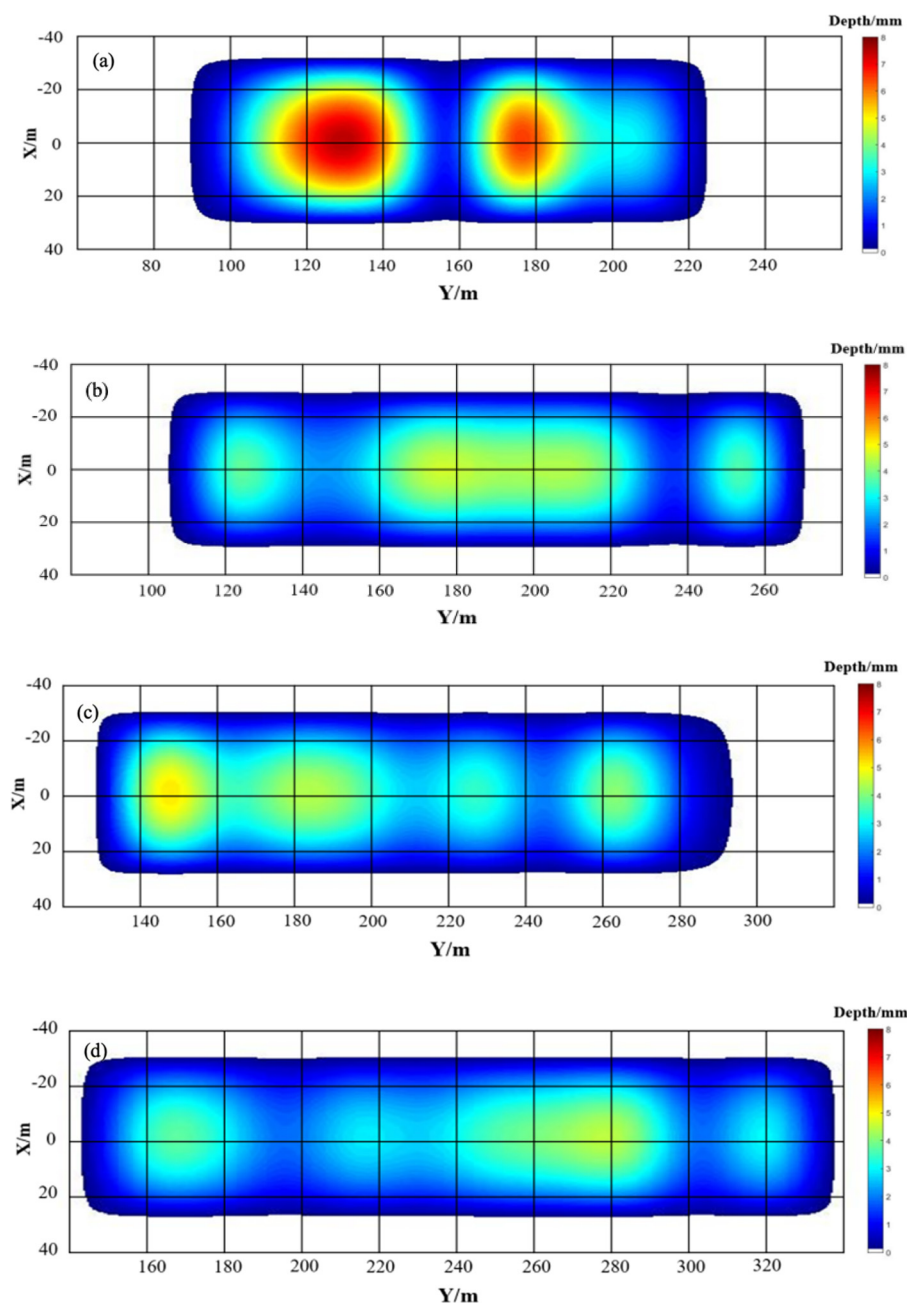


FIG. 22. Water-dropping range and density result: (a) drop height 50 m; (b) drop height 100 m; (c) drop height 150 m; and (d) drop height 200 m.

Figure 24 illustrates the distribution of water-dropping density and the corresponding fire-extinguishing areas under different discharge volumes. Areas with a water drop density greater than 0.2 l/m^2 are defined as the water coverage area (cyan), whereas areas with a density greater than 2 l/m^2 are defined as the effective fire-extinguishing area (red). The fire-extinguishing area calculated using the heat-exchange model is shown in green. The heat-exchange-based results (green) show smaller, but more concentrated, fire-suppression areas than the empirical criteria. This difference reflects the influence

of combustible calorific value and demonstrates that the proposed model can adapt to various materials for more accurate performance prediction.

V. CONCLUSIONS AND PROSPECTS

This paper proposed and validated a coupled VOF-enhanced-DPM algorithm to efficiently simulate droplet trajectory, breakup, and distribution in firefighting aircraft water-dropping. Compared with the VOF method, this algorithm significantly reduces the computational

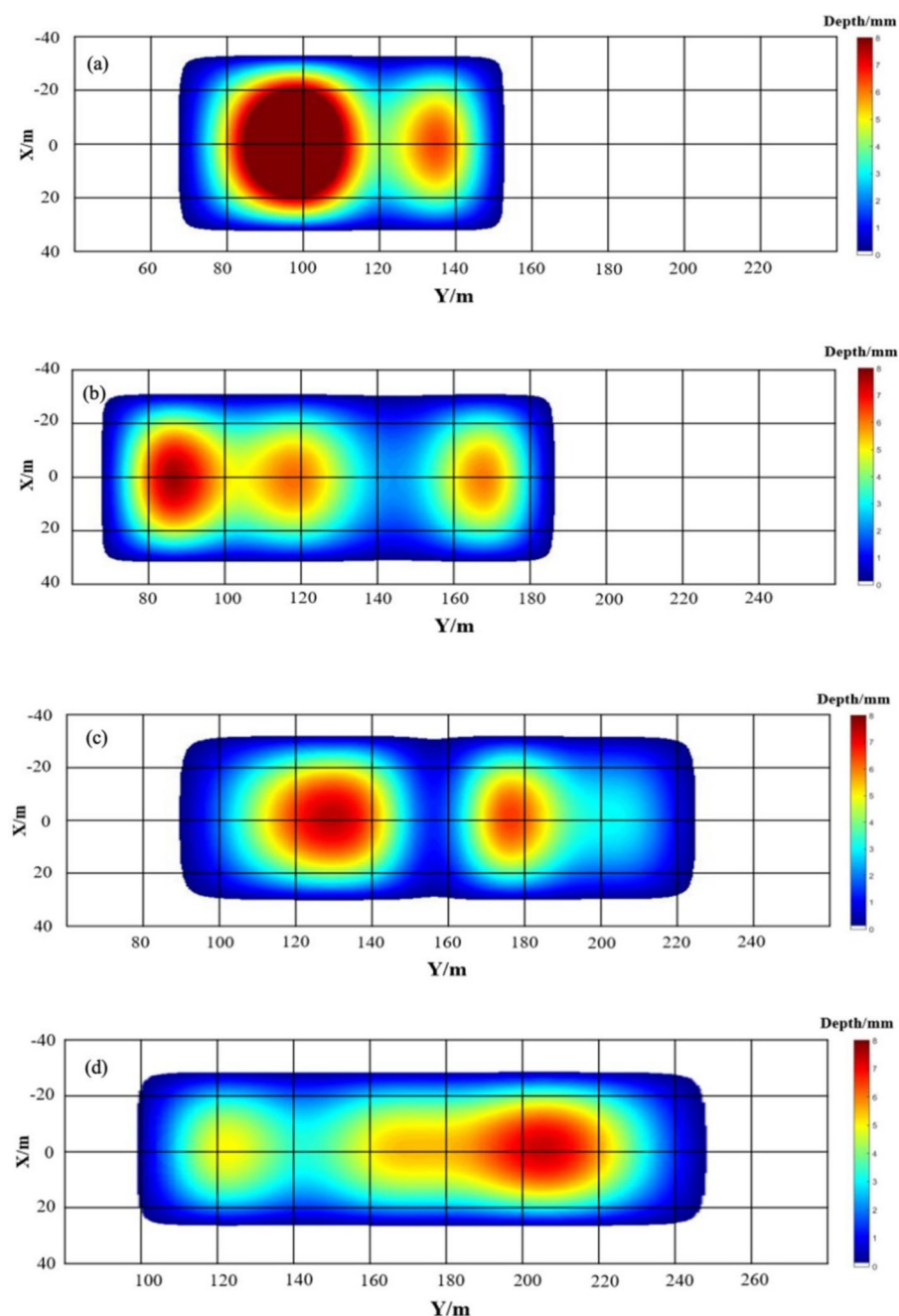


FIG. 23. Water-dropping range and density result: (a) flight velocity 150 km/h; (b) flight velocity 200 km/h; (c) flight velocity 230 km/h; and (d) flight velocity 250 km/h.

cost while maintaining accurate interface tracking. Although enhanced-DPM simplifies the small-scale breakup details, it preserves the accuracy of the overall droplet distribution.

The algorithm was further applied to large-scale simulations of the AG600 firefighting aircraft. The results indicate that discharge volumes, heights, and velocity significantly affect droplet distribution. Larger water volumes increase droplet density and concentrate the distribution, while greater heights expand the coverage area but reduce

density. Flight velocity influences both the range and uniformity of the droplet field. Moreover, firefighting performance evaluations show that empirical criteria tend to overestimate effectiveness, whereas heat-exchange models provide a more realistic assessment by accounting for combustible calorific values. This offers practical guidance for optimizing water-dropping strategies.

In the future, the algorithm will extend to more complex fire environments, including the coupling effects of non-uniform turbulent

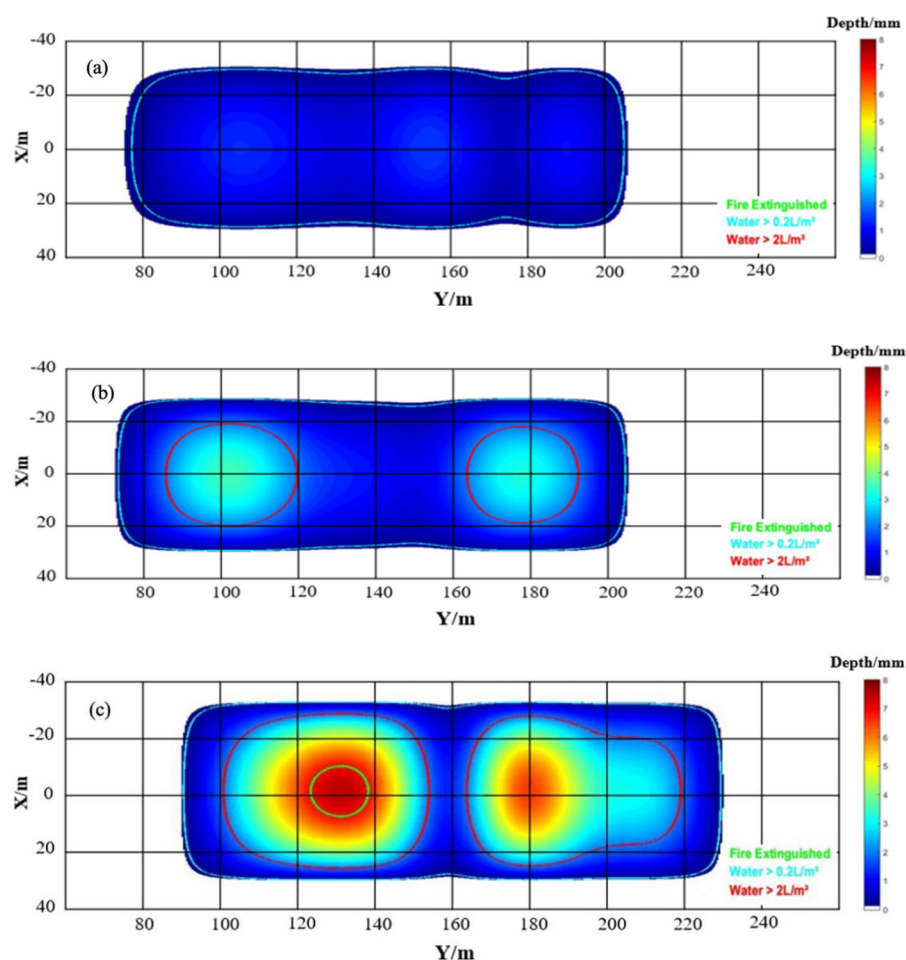


FIG. 24. Comparison of fire-extinguishing areas for AG600 under different criteria: (a) water volume 3330 l; (b) water volume 6030 l; and (c) water volume 12 000 l.

wind fields, varying extinguishing agent properties, and multiple water-dropping operations.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yumeng Shi: Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal). **Qiuze Yang:** Investigation (equal); Validation (equal). **Xinzhe**

Chang: Formal analysis (equal); Investigation (equal); Writing – review & editing (lead). **Jiyuan Ye:** Formal analysis (equal); Validation (equal). **Fei Xu:** Conceptualization (equal); Methodology (equal); Resources (equal); Supervision (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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